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Agenda

* Problems on Bayes theorem

** Assignments

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① A and B are 2 independent Events, where it is known that $P(A \cup B) = 0.5$ and $P(A) = 0.3$, what is $P(B)$?

** $P(A \cap B) = P(A) * P(B)$

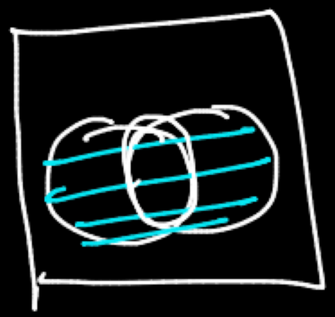
$P(A|B) = P(A)$

$\downarrow$

$\frac{P(A \cap B)}{P(B)} = P(A)$

$P(A \cap B) = P(A) * P(B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$



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$$P(A \cup B) = P(A) + P(B) - \boxed{P(A \cap B)}$$

$$P(A \cup B) = P(A) + P(B) - P(A) * P(B)$$

$$0.5 = 0.3 + P(B) - 0.3 * P(B)$$

$$0.2 = P(B)\,(1 - 0.3)$$

$$0.2 = P(B)\,(0.7)$$

$$P(B) = \frac{0.2}{0.7} = \frac{2}{7} = 0.28\ldots$$

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Bayes theorem

$$P(A \mid B) = \frac{P(B \mid A) * P(A)}{P(B)}$$

Quiz #1

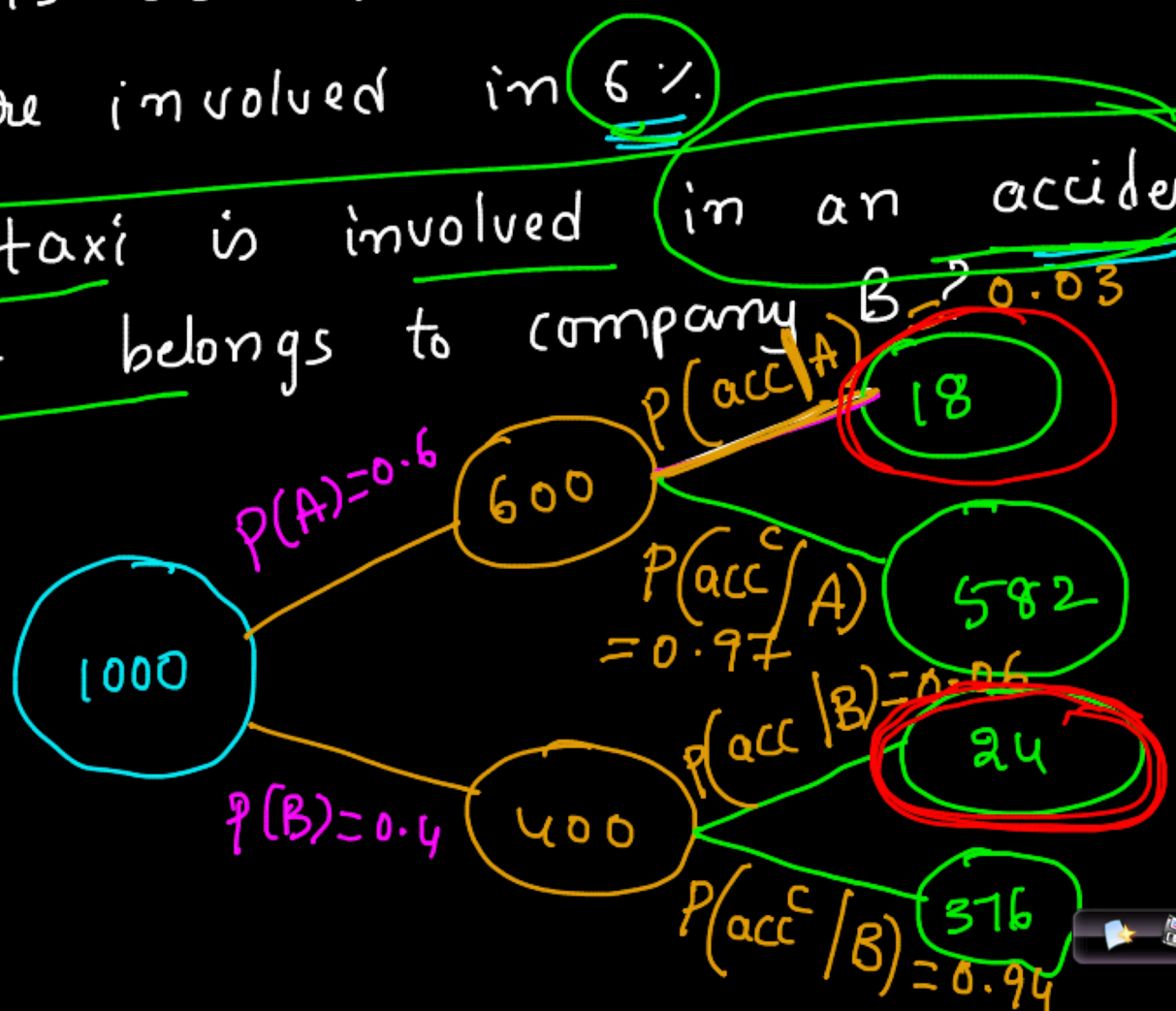
2 taxi companies A & B

- ✓ A has 60 % of taxis in the city
- ✓ B has 40 %

⇒ A's taxis are involved in accidents 3% of the time,

⇒ B's are involved in 6 %.

If a taxi is involved in an accident, what is the probability that it belongs to company B ? 0.03



$$P(B \mid acc) = \frac{24}{18+24} = \frac{24}{42} = 0.57$$

$$P(B \mid acc) = \frac{P(acc \mid B) * P(B)}{P(acc)}$$

$P(acc)$ → total law of probability

$$P(acc) = P(acc \mid A) * P(A) + P(acc \mid B) * P(B)$$

$$= 0.6 * 0.03 + 0.06 * 0.4$$

$$= 0.18 + 0.24$$

$$= \frac{0.24}{0.42} = \frac{24}{42}$$

$$= 0.57$$

Quiz #2

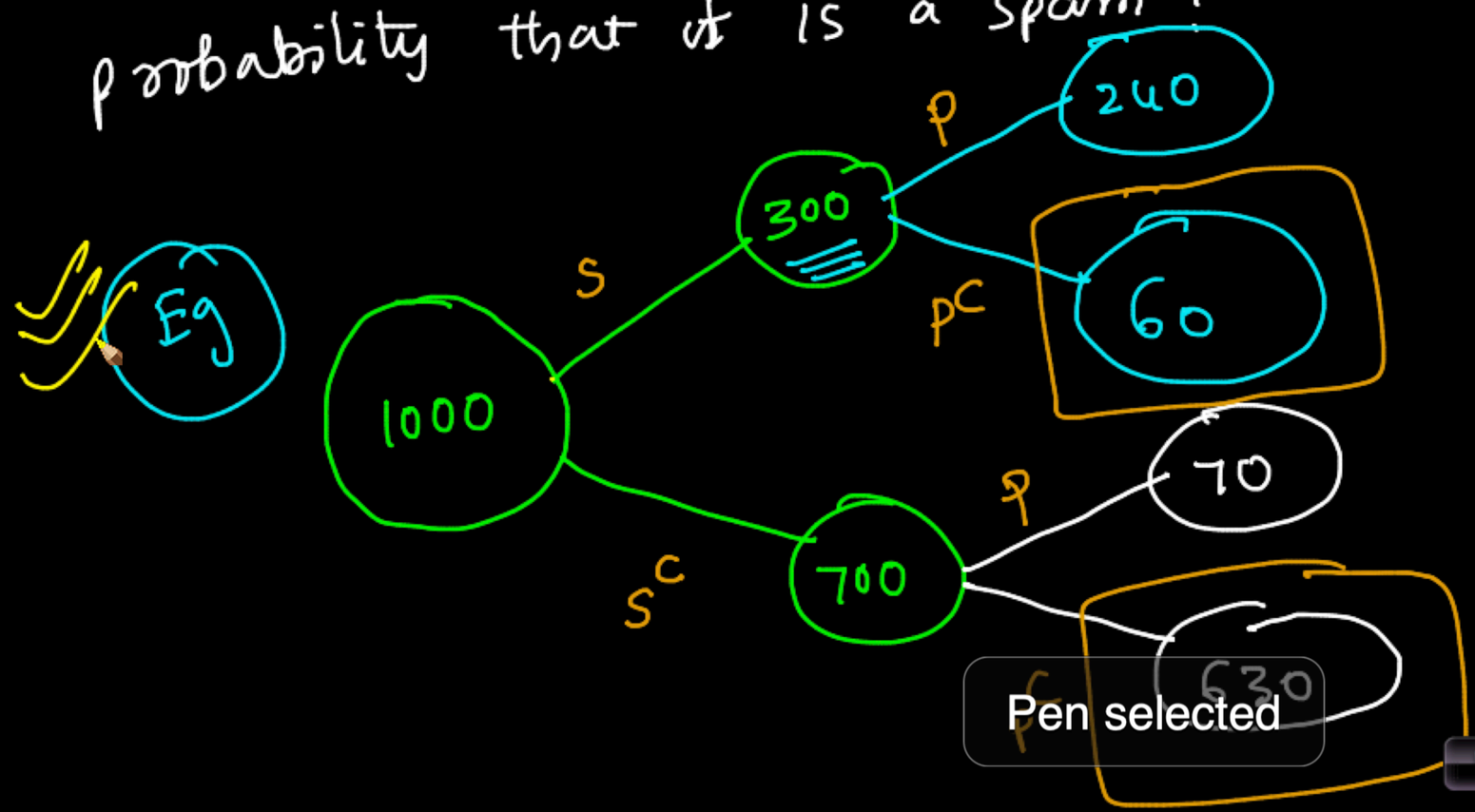
30 % of emails are spam

70 % are not spam

⇒ 80 % of spam emails contains word "purchase"

10 % of non-spam " " " "

A new email does not have a word purchase. What is the probability that it is a spam?



$$P\left(S \mid P^c\right)$$

$$\frac{60}{690}$$

$$= 0.08 \text{ //}$$

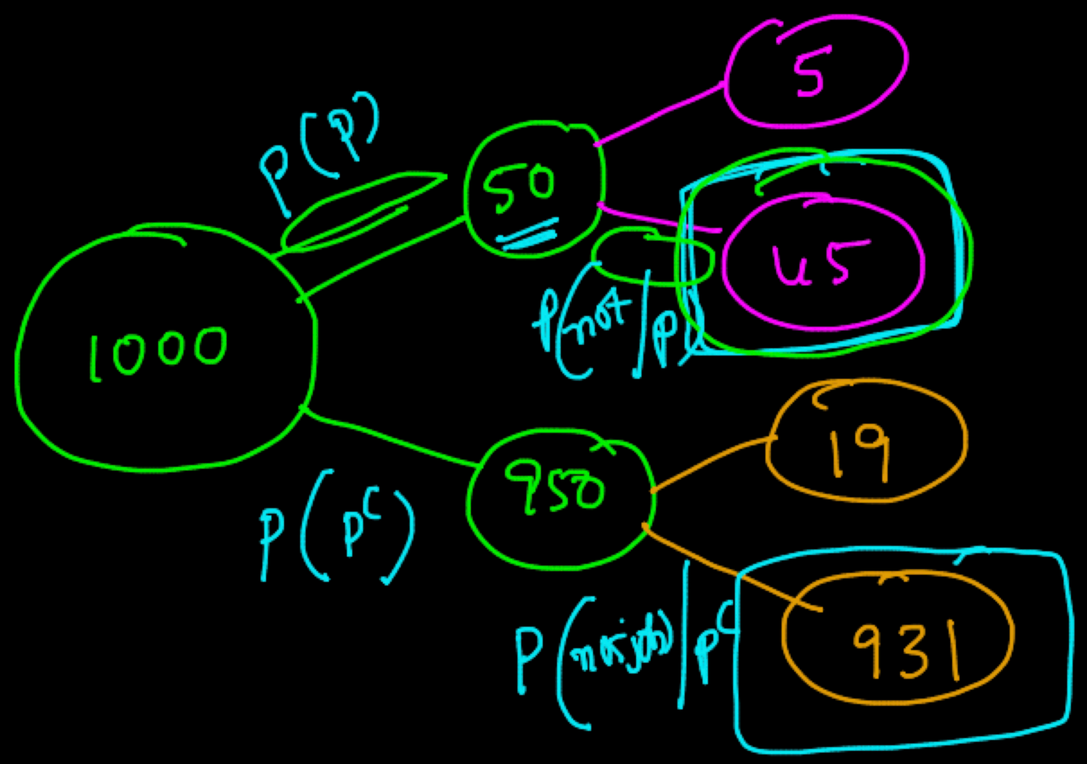
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Quiz #3

5% LinkedIn Users are premium

10% of premium users are seeking new job.

2% of non-premium users are seeking new jobs. A randomly choosen person is not seeking new jobs. What is the probability that he is a premium user?



$$= \frac{45}{976} = 0.046$$

$$P(\text{prem} \mid \text{not job})$$

Directly given

1) $P(prem) = 0.05$

2) $P(\text{not } prem) = 0.95$

3) $P(job \mid prem) = 0.1 \Rightarrow P(\text{no job} \mid prem) = 0.9$

4) $P(job \mid \text{not } prem) = 0.02$

↳ $P(\text{no job} \mid \text{not } prem) = 0.98$

Not directly given ⇒ Bayes Theorem

$$P(prem \mid job) = \frac{P(job \mid prem) * P(prem)}{P(job)}$$

$$P(prem \mid \overline{not\ job}) = \frac{P(\overline{not\ job} \mid prem) * P(prem)}{P(\overline{not\ job})}$$

$$P(\overline{not\ prem} \mid job) = \frac{P(job \mid \overline{not\ prem}) * P(\overline{not\ prem})}{P(job)}$$

$$P(\overline{not\ prem} \mid \overline{not\ job}) = \frac{P(\overline{not\ job} \mid \overline{not\ prem}) * P(\overline{not\ prem})}{P(\overline{not\ job})}$$

$$= \frac{0.9 * 0.05}{0.05 * 0.9 + 0.95 * 0.98}$$ ✓

$$P(\text{not job}) = P(\text{prem}) * P(\text{not job} \mid \text{prem}) +$$

$$P(\text{not prem}) * P(\text{not job} \mid \text{not prem})$$

$$= \boxed{0.05 * 0.9 + 0.95 * 0.98}$$

Quiz

A website shows 2 types of ads. 60% of the visitors see type A ads, 40% of the visitors see type B ads. The CTR for A is 5%, and for B is 3%. A visitor to the website does not click the ad. What is the probability that he saw type A ad?

$P(A) = 0.6$ $\quad P(B) = 0.4$

$P(CTR \mid A) = 0.05 \Rightarrow P(CTR^c \mid A) = 0.95$

$P(CTR \mid B) = 0.03 \hookrightarrow P(CTR^c \mid B) = 0.97$

$P(A \mid \text{not click}) =$

$$P(A \mid C^c) = \frac{P(C^c \mid A) * P(A)}{P(C^c)}$$

$$= \frac{0.95 * 0.6}{0.95 * 0.6 + 0.97 * 0.4} = 0.59$$

$$P(C^c) = P(C^c \mid A) * P(A) + P(C^c \mid B) * P(B)$$

$$= 0.95 * 0.6 + 0.97 * 0.4$$

Suppose 5% of men and 0.25% of women are color blind. A random color-blind person is chosen. What is the probability of this person being male?

Assume there are equal no of men and women overall

$P(M) = 0.5$

$P(W) = 0.5$

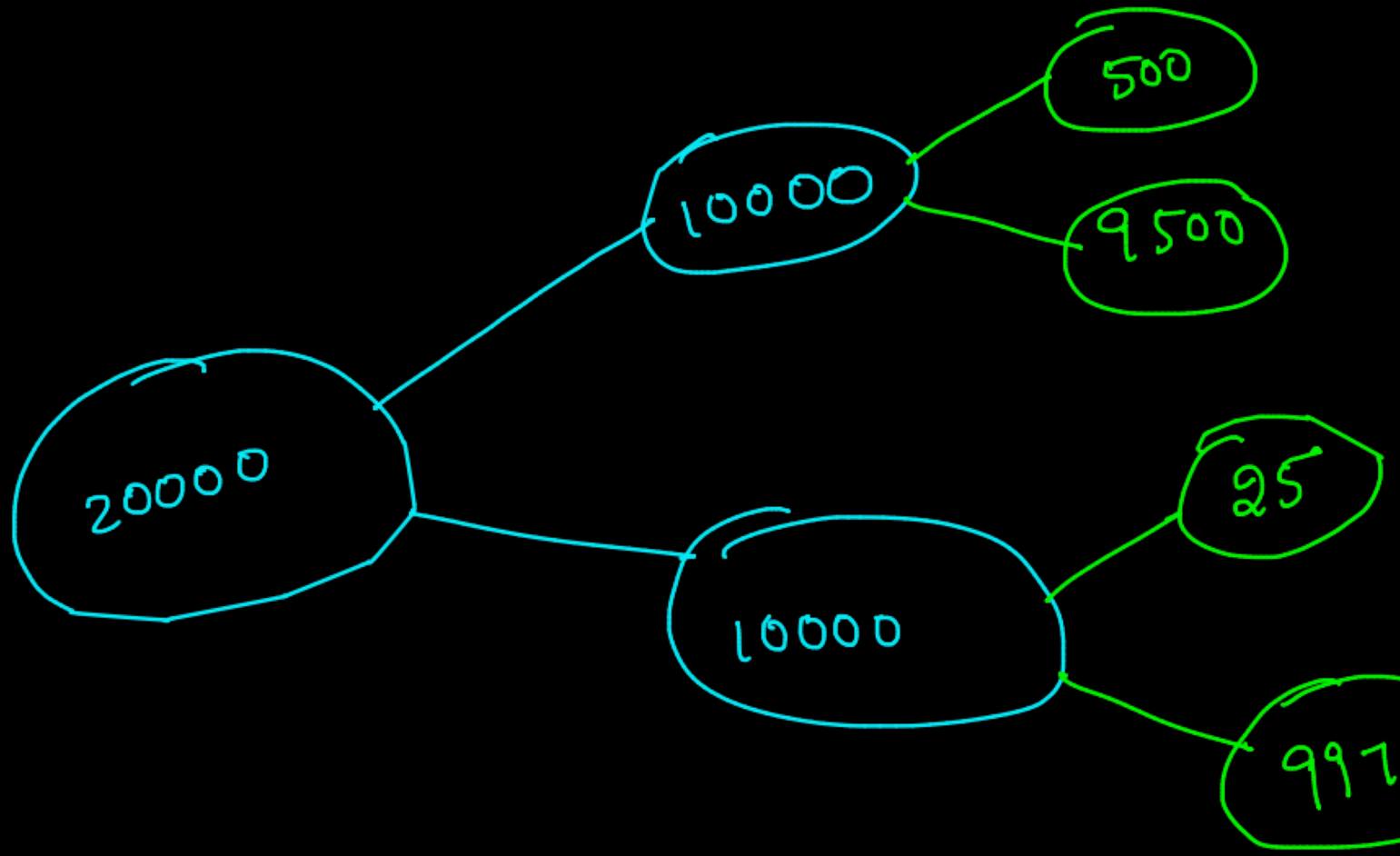
$P(CB \mid m) = 0.05$

$P(CB \mid w) = 0.0025$

$P(M \mid CB) = \frac{P(CB \mid M) * P(M)}{P(CB)}$

$= \frac{0.05 * 0.5}{0.05 * 0.5 + 0.0025 * 0.5}$

$= 0.95$



$$\frac{500}{525} = 0.95$$

⇒ A gambler has in his pocket fair coin and two headed coin. He selects one of the coins at random, and when he flips it, it shows heads. What is the probability that it is a fair coin?

$P(F) = 0.5$

$P(B) = 0.5$

$P(H|F) = 0.5$

$P(H|B) = 1$

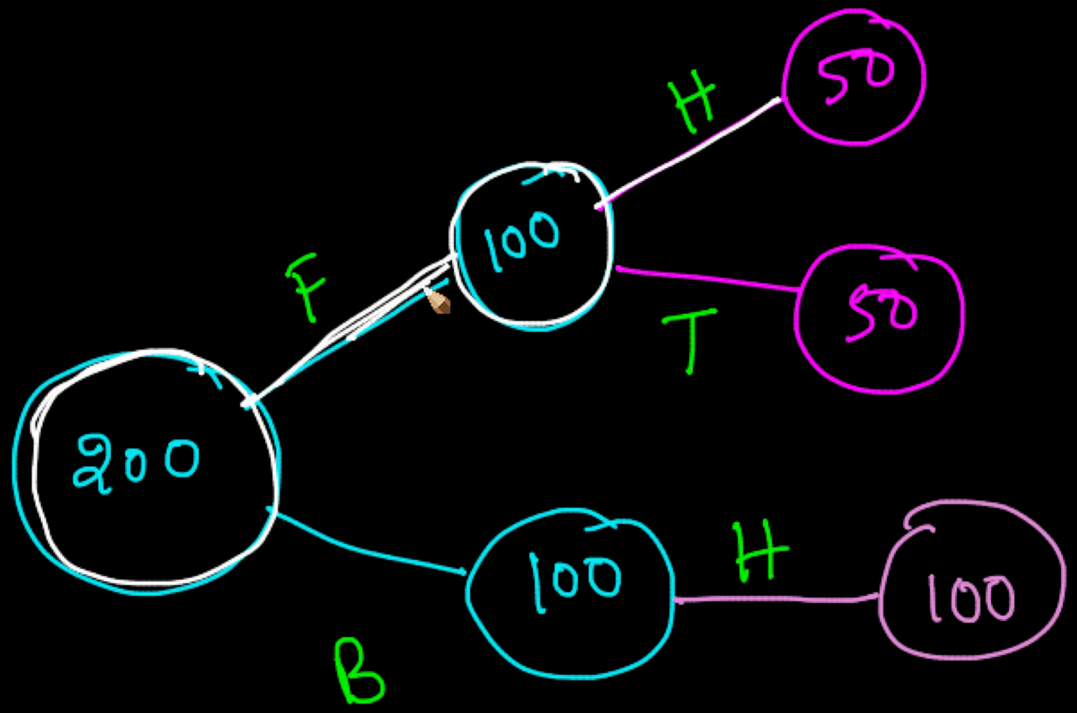
$P(F|H) = \frac{P(H|F) * P(F)}{P(H)}$

$= \frac{0.5}{1.5} = \frac{5}{15} = 1/3$

$P(H) = P(H|F) * P(F) + P(H|B) * P(B)$

$0.5 * 0.5 + 1 * 0.5$

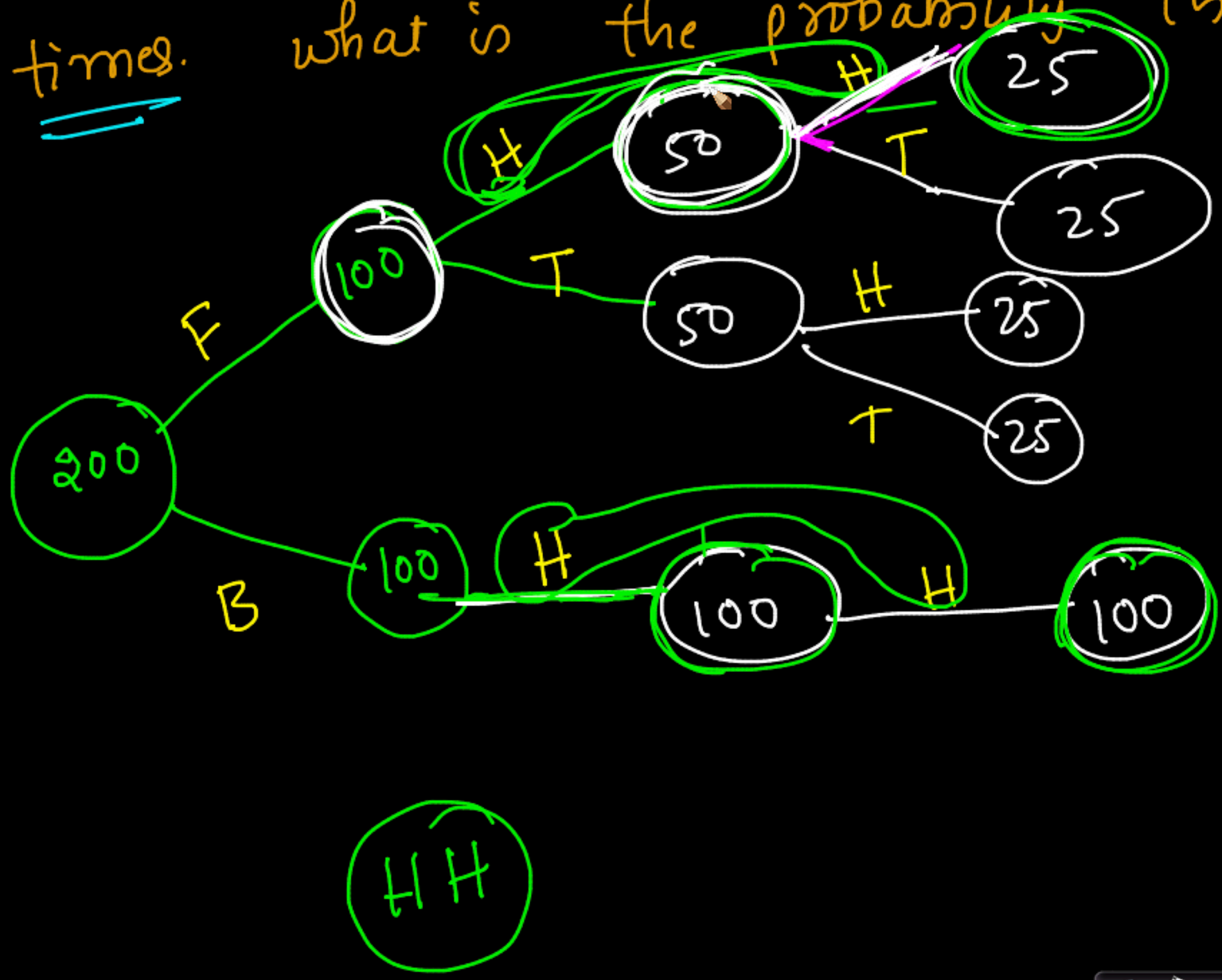
$\frac{0.5 * 0.5}{0.5 * 0.5 + 1 * 0.5}$



$$P(F|H) = \frac{50}{150}$$

$$= \frac{5}{15} = \boxed{1/3}$$

=> He flip it twice. It shows heads both The times. what is the probability that it is a fair coin?



$$\frac{25}{125} = \frac{1}{5}$$

flip it 3 times ~~thrice~~. He get $\{H\ H\ T\}$. What is the probability that it is a fair coin?



⇒ A family has 2 children, atleast one of them is a girl what is the probability that both are girls?

$S = \{BB, BG, GB, GG\}$

A : both are girls

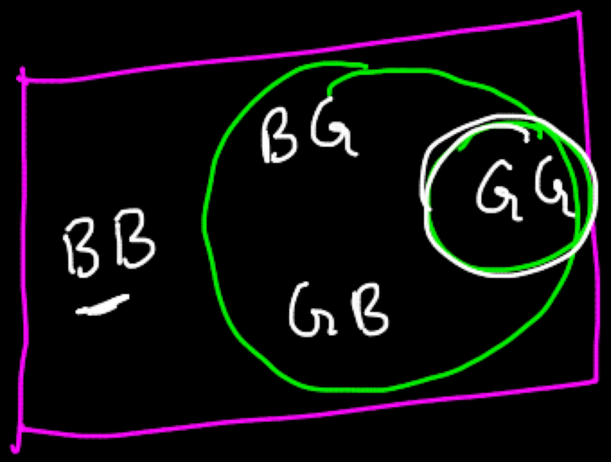
$A = \{GG\}$

$P(A) = 1/4$

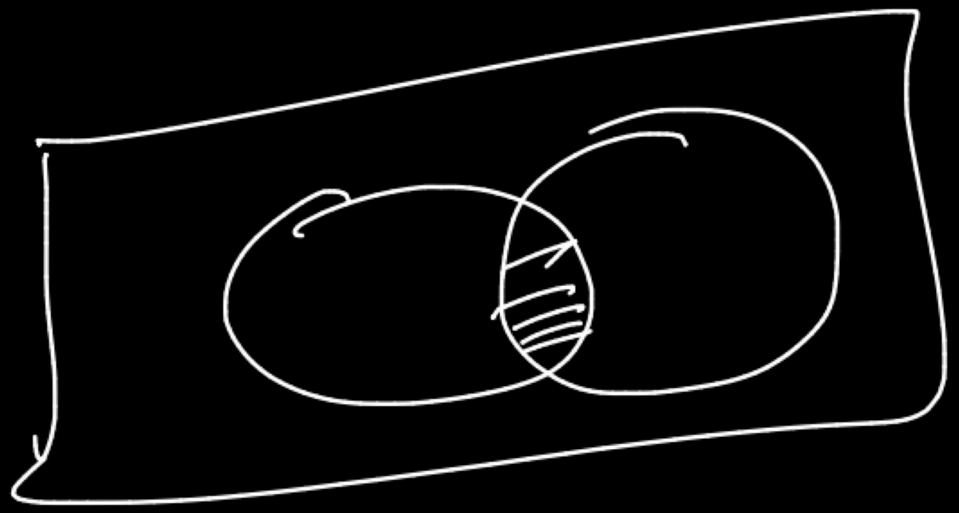
B : atleas one of them girl

$B = \{BG, GB, GG\}$

$P(B) = 3/4$



$P(A \cap B) = 1/4$



{ }

$(A \cap B) =$

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$$P(A|B) = \frac{P(A \cap B)}{P(B)} \neq 0 \neq P(A)$$

$$P(A|B) = P(A)$$